FAMA: Feature-Space Adjoint Attribution for Explaining Negative Transfer in Gradient-Based Meta-Learning

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Abstract

Gradient-based meta-learning adapts a model to a new task from a handful of labeled examples, yet adaptation can hurt, and no existing method explains why on a given task. We introduce the adaptation gain ΔM, a signed and differentiable measure of whether backbone adaptation helps relative to a feature-reuse counterfactual, and FAMA, a post-hoc explainer that computes the exact derivative of ΔM with respect to a spatial mask on the support features. An adjoint recursion over the inner loop computes this derivative at the cost of one double-backward pass per step, and an integrated variant decomposes ΔM exactly across support regions. FAMA unifies trajectory-based data attribution, such as TracIn and SGD-influence, with spatial saliency. Because fast adaptation can be re-run cheaply, we also build protocols with counterfactual ground truth, including re-adaptation and a planted-feature benchmark. Across **[N]** tasks and **[K]** backbones, FAMA **[headline result]**, reveals contextual bias as a driver of negative transfer, and guides support-set edits that recover **[X]%** of the lost accuracy.

1. Introduction

Large pretrained foundation models have become the default starting point for downstream tasks (Bommasani et al., 2021). Yet they are no panacea in data-scarce domains such as medical imaging, where rare diseases offer only a few labeled cases (Zhang & Metaxas, 2024). There, and in domains such as remote sensing, a model must adapt to each new task from a handful of labeled examples (Guo et al., 2020; Silva-Rodríguez et al., 2025). Updating the weights on these examples can outperform in-context learning (Gozeten et al., 2025; Ghosh et al., 2026), particularly on structurally novel tasks (Akyürek et al., 2025). Meta-learning (Hospedales et al., 2022), epitomized by MAML (Finn et al., 2017), makes such updates effective by meta-training an initialization from which a few gradient steps suffice. Learning to adapt now reappears at foundation-model scale, in test-time training that meta-learns its initialization (Sun et al., 2025; Tandon et al., 2025) and in language models that learn to generate their own weight updates (Zweiger et al., 2025).

Adaptation, however, does not always help. Letting the backbone adapt on a noisy, hard, or shifted support set can raise the query loss, a phenomenon known as negative transfer (Rosenstein et al., 2005; Wang et al., 2019). Under distribution shift, fine-tuning a whole pretrained network can underperform tuning only its head or a few layers (Kumar et al., 2022; Lee et al., 2023). In meta-reinforcement learning, the policy that MAML adapts can even underperform its initialization (Deleu & Bengio, 2018). Prior work mitigates negative transfer or its likely causes, such as confounding pretrained knowledge (Yue et al., 2020), noisy or hard tasks (Yao et al., 2021; Collins et al., 2020), and spurious features (Parast et al., 2026). Analyses of adaptation, in turn, describe the average task (Raghu et al., 2020; Oh et al., 2021). None explains, for a given task, whether backbone adaptation helped and which support-set features drove the outcome.

Existing explanation methods capture only half of this picture. Saliency maps such as Grad-CAM (Selvaraju et al., 2017) localize evidence, but at a single parameter state rather than along the adaptation trajectory. Training-data attribution along the trajectory (Pruthi et al., 2020; Hara et al., 2019) scores whole examples rather than the features within them. Influence functions can score the features of a training input, but only at a converged model (Koh & Liang, 2017), whereas fast adaptation stops after a few steps. Meta-learning variants of these methods also require the meta-training data (Mitsuka et al., 2026; Ren et al., 2026).

Our key observation is that, at meta-test time, the support set is both the training data of the adaptation trajectory and the input we wish to explain. We first define the **adaptation gain** ΔM, whose sign tells whether backbone adaptation helped or hurt relative to a feature-reuse counterfactual. We

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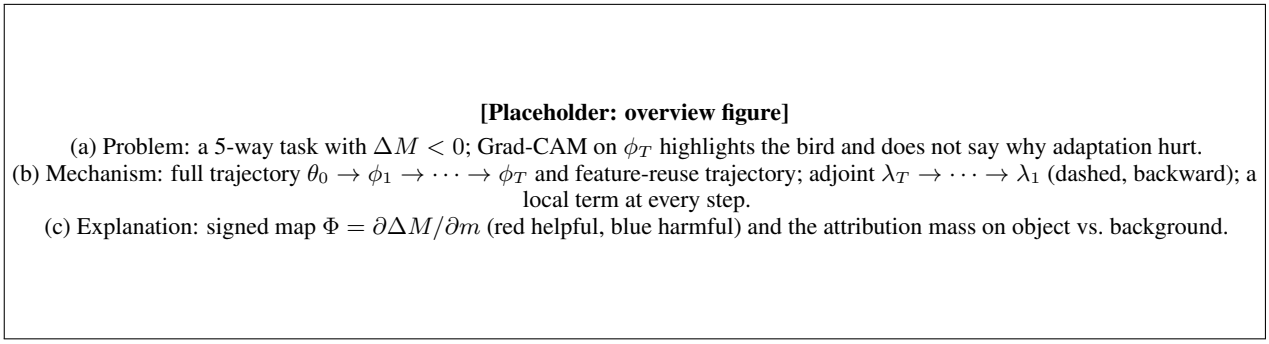


Figure 1. Overview of FAMA. (a) On a task where backbone adaptation hurts, an end-state explanation does not reveal why. (b) FAMA runs the full and the feature-reuse inner loops and carries the sensitivity of the query risk backward with the exact adjoint. (c) The resulting signed map attributes the adaptation gain to support regions, here exposing harm on the background. **[Placeholder.]**

then introduce **FAMA**, which differentiates ΔM with respect to a spatial mask on the support features through a discrete adjoint recursion over the inner loop, without any meta-training data (Figure 1). Instance-level trajectory attribution methods arise as special cases of the same template, so FAMA is their feature-level counterpart (Section 3).

Because an inner loop takes only a few steps, it can be rerun after every intervention on the support set. This makes counterfactual ground truth affordable, and we build our faithfulness protocols on it (Section 4).

Across **[N]** tasks, **[K]** backbones, and **[D]** domains, FAMA **[headline result vs. the strongest baseline]**. It reveals *contextual bias* as a causal driver of negative transfer **[result]**, and editing the regions it flags recovers **[X]%** of the lost accuracy. In summary, we make the following contributions:

- **Adaptation gain.** We define ΔM, a differentiable, sign-aware measure of whether backbone adaptation helps or hurts on a task, and characterize it to leading order (Section 3.1 and Proposition 3.2).
- **FAMA.** We introduce a post-hoc explainer that computes the exact derivative of ΔM with respect to support-set features through the whole trajectory, and an integrated variant that decomposes ΔM exactly across support regions (Section 3). It unifies trajectory-based data attribution with spatial saliency.
- **Ground-truth evaluation.** We propose protocols for trajectory explanations, including a bidirectional deletion test, counterfactual re-adaptation, and a planted-feature benchmark (Section 4).
- **Findings.** We show that FAMA outperforms static and instance-level baselines and extends beyond MAML to few-shot LoRA adaptation of CLIP (Section 5).

2. Preliminaries and Related Work

2.1. Meta-learning and MAML

Meta-learning, or learning to learn (Schmidhuber, 1987; Bengio et al., 1991; Hospedales et al., 2022), trains a learner across a distribution of tasks so that it adapts quickly to new ones. In N-way K-shot classification, each task $\mathcal{T}_i$ provides a labeled *support set* $S_i = \{(x_k, y_k)\}_{k=1}^{NK}$ for adaptation and a *query set* Q_i for evaluation. MAML (Finn et al., 2017) meta-trains a shared initialization $\theta_0 \in \mathbb{R}^P$ and adapts it to $\mathcal{T}_i$ with T gradient steps on the mean support loss $\mathcal{L}(\phi, S_i)$, starting from $\phi_0 = \theta_0$,

$$\phi_t = \phi_{t-1} - \alpha \nabla_\phi \mathcal{L}(\phi_{t-1}, S_i), \qquad t = 1, \ldots, T. \quad (1)$$

The adapted model is thus the endpoint of the trajectory $\theta_0 \to \phi_1 \to \cdots \to \phi_T$, along which we write $H_t := \nabla^2_\phi \mathcal{L}(\phi_t, S_i)$ for the support-loss Hessian. Every state splits into a backbone ϕ_t^b, which maps an image x to a feature map $A = f_{\phi_t^b}(x) \in \mathbb{R}^{C \times \Omega}$ on a spatial grid Ω, and a head ϕ_t^h that classifies from A. We measure performance by the query risk $R(\phi) := \mathbb{E}_{Q \sim \mathcal{T}_i}[\mathcal{L}(\phi, Q)]$, the expected query loss of the task.

2.2. Negative transfer and feature reuse

Negative transfer occurs when knowledge carried over from a source hurts performance on a target task (Rosenstein et al., 2005; Wang et al., 2019). In MAML, the inner loop itself can be harmful (Deleu & Bengio, 2018). A single shared initialization also struggles when tasks are heterogeneous, which motivates priors tailored to task clusters, task relations, or modalities (Yao et al., 2019; 2020; Chen & Zhang, 2021). Bayesian TAML instead learns, for each task, how much to rely on meta-knowledge rather than task-specific learning (Lee et al., 2020). Other work models pretrained knowledge as a confounder (Yue et al., 2020), reweights tasks by their noise or difficulty (Yao et al., 2021; Collins et al., 2020), or edits support embeddings to counter spurious correlations (Parast et al., 2026). Raghu et al. (2020)

find that much of MAML's benefit comes from reusing backbone features while only the head adapts, whereas Oh et al. (2021) argue for representation change. We ask which view holds on each task. Full adaptation is compared with a *feature-reuse* branch (Section 3.1), which freezes the backbone at θ_0^b, as in the test-time experiment of Raghu et al. (2020).

2.3. Explaining predictions and training

Saliency methods attribute a prediction of a fixed model to its input, through gradients (Simonyan et al., 2014), path integrals (Sundararajan et al., 2017), Shapley values (Lundberg & Lee, 2017), or gradient-weighted feature maps (Selvaraju et al., 2017). Training-data attribution instead asks how training examples shaped a model. Influence functions (Koh & Liang, 2017) linearize the response of a converged model, and TracIn (Pruthi et al., 2020) sums gradient alignments across checkpoints. SGD-influence (Hara et al., 2019) propagates the effect of an example backward through the steps of SGD. The trajectory-based methods score whole examples rather than regions within them. Influence functions can also score the features of a training input, but only at a converged model (Koh & Liang, 2017). Explanations for meta-learning trace behavior to the meta-training tasks (Mitsuka et al., 2026; Ren et al., 2026), which requires that corpus at explanation time. For few-shot models, explanations propagate relevance through a metric-based classifier (Sun et al., 2021) or come from concepts built into the learner (Cao et al., 2021). Neither explains gradient-based adaptation from θ_0 to ϕ_T. Test-time training raises the same question of whether an update helps (Sun et al., 2020; Wakayama, 2026), but it does not attribute the answer to input features. Appendix D extends this discussion.

Gradient saliency and training-data attribution both measure a first-order sensitivity of an output or a loss to a perturbation. Saliency perturbs the input of a fixed model, whereas training-data attribution perturbs the data that shaped the model. The latter methods differ in what they perturb, from whole examples to the features of an input, and in how they carry its effect along the training trajectory. Section 3.4 makes this precise. FAMA attributes the effect of adaptation to regions of the support images, follows the whole trajectory exactly, and needs only the support set. No method above has all three properties.

3. FAMA: Feature-Space Adjoint Attribution

FAMA answers two questions about a task $\mathcal{T}_i$ (Figure 1). Did backbone adaptation help or hurt, and which regions of the support images are responsible? We answer the first with a scalar, the adaptation gain (Section 3.1). We answer the second with the exact derivative of that scalar with respect to a spatial mask on the support features, computed by an adjoint pass over the inner loop (Sections 3.2 and 3.3). Section 3.4 relates the result to prior attribution methods, and Section 3.5 gives the algorithm and its cost.

3.1. Adaptation gain

Let $\phi_t^f = (\theta_0^b, \phi_t^{f,h})$ denote the feature-reuse trajectory, which starts from θ_0 but updates only the head, $\phi_t^{f,h} = \phi_{t-1}^{f,h} - \alpha \nabla_{\phi^h} \mathcal{L}(\phi_{t-1}^f, S_i)$. We define the *adaptation gain* as the gap in query risk between the endpoints of the two branches,

$$\Delta M := R(\phi_T^f) - R(\phi_T). \tag{2}$$

A positive ΔM means that letting the backbone move under S_i beats feature reuse, a negative one means that it actively hurts, and $\Delta M \approx 0$ recovers the feature-reuse regime of Raghu et al. (2020). Unlike CCA (Raghu et al., 2017) or CKA (Kornblith et al., 2019), which measure how much the representation changes, ΔM is signed and measures whether the change helps on the query set. It is also differentiable in the support set, which makes it attributable. We estimate the expectation in R by a stratified bootstrap over query sets and evaluate both branches on the same resamples, which reduces the variance of their difference (Appendix C.1). In classification, $\mathcal{L}$ and R are cross-entropies. For a non-differentiable metric such as accuracy or Dice, R can be any differentiable surrogate, and we report how often its sign agrees with that of the metric.

Assumption 3.1 (Fast adaptation)**.** Along both trajectories, the support gradient $\nabla_\phi \mathcal{L}(\cdot, S_i)$ has norm at most γ and is κ-Lipschitz. The Hessian of R is κ_H-Lipschitz on a neighborhood of θ_0 that contains both trajectories. Moreover, $T\alpha\kappa \le 1$, and the displacement bound $T\alpha\gamma$ is small, so that a second-order expansion of R around θ_0 is accurate (cf. the local quadratic analysis of fast adaptation by Grant et al., 2018).

Proposition 3.2 (Backbone dominance)**.** *Let $g_\bullet$ and $H^R_{\bullet\bullet}$ be the blocks of the gradient and Hessian of R at θ_0, and let $\Delta^b := \phi_T^b - \theta_0^b$ and $\Delta^h := \phi_T^h - \theta_0^h$. Under Assumption 3.1,*

$$\Delta M = -C_b + O\big((T\alpha)^2\big) \tag{3}$$

$$= T\alpha \left\langle g_b, \nabla_{\phi^b} \mathcal{L}(\theta_0, S_i) \right\rangle + O\big((T\alpha)^2\big), \tag{4}$$

where $C_b := g_b^\top \Delta^b + \frac{1}{2}\Delta^{b\top} H^R_{bb} \Delta^b + \Delta^{b\top} H^R_{bh} \Delta^h = O(T\alpha)$ collects every term that involves the backbone displacement. For $T = 1$, the remainder in (3) is $O(\alpha^3)$. The constants in $O(\cdot)$ depend only on γ, κ, κ_H, and the derivatives of R at θ_0.

Proposition 3.2 (proof in Appendix A.2) justifies reading ΔM as a property of the backbone's displacement. By (4), backbone adaptation helps to leading order when the backbone gradients of the support loss and the query risk align. Negative transfer then corresponds to a conflict between the two gradients. The result is local, however, and it does not

assign the sign of every finite-step ΔM to the backbone alone.

3.2. Exact adjoint of the inner loop

Each inner-loop step has the Jacobian $\partial\phi_t/\partial\phi_{t-1} = I - \alpha H_{t-1}$, a dense $P \times P$ matrix that we never form. Instead, we propagate the sensitivity of the query risk backward, for $t = T, \dots, 1$, as the discrete adjoint, or co-state, of the inner loop (Pontryagin et al., 1962),

$$\lambda_T := \nabla_\phi R(\phi_T), \qquad \lambda_{t-1} = \lambda_t - \alpha H_{t-1}\lambda_t. \quad (5)$$

Lemma 3.3 (Exact adjoint). *The recursion (5) gives $\lambda_t = \partial R(\phi_T)/\partial\phi_t$ for every t, with no approximation beyond (1).*

Each step costs one Hessian-vector product, computed exactly without materializing H_{t-1} (Pearlmutter, 1994), here by double backpropagation, and Appendix B.1 bounds how sampling and rounding errors propagate through it. The recursion extends to any differentiable update $\phi_t = U(\phi_{t-1}, S_i)$ as $\lambda_{t-1} = (\partial U/\partial\phi_{t-1})^\top \lambda_t$ (Appendix B.2). Per-parameter learning rates (Li et al., 2017), for instance, give $\lambda_{t-1} = \lambda_t - H_{t-1}(\alpha \odot \lambda_t)$, and low-rank adapters (Hu et al., 2022) restrict ϕ to the adapter parameters. In the feature-reuse branch, only the head moves, so the recursion runs over the head parameters with the head block of the Hessian at ϕ^f_{t-1}. Li et al. (2023) use a continuous-time adjoint of gradient flow to make meta-*training* cheaper. We use the discrete adjoint instead to explain an already trained model at meta-test time.

3.3. Feature-space attribution

To localize ΔM, we attach a spatial mask $m_k \in \mathbb{R}^\Omega$ to the backbone output of each support image. The mask replaces the feature map A_k by $m_k \odot A_k$, broadcast over channels, inside the support loss $\mathcal{L}(\phi, S_i; m)$ of both branches and at every inner-loop step. Query evaluation is left unmasked. This defines a function $\Delta M(m)$ with $\Delta M(\mathbf{1}) = \Delta M$, in which lowering m_k over a region suppresses that region of x_k during adaptation. We attribute at the feature map rather than at the pixels for two reasons. Like the last convolutional layer used by Grad-CAM, the backbone output balances high-level semantics with spatial detail (Selvaraju et al., 2017), and pixel gradients are visually noisy (Smilkov et al., 2017).

Proposition 3.4 (Exact feature attribution). *Let λ_t and λ^f_t be the adjoints (5) of the two branches. Define the local terms*

$$G_{t,k} := \frac{\partial}{\partial m_k}\Big[\lambda_t^\top \nabla_\phi \mathcal{L}(\phi_{t-1}, S_i; m)\Big]_{m=\mathbf{1}} \in \mathbb{R}^\Omega, \quad (6)$$

with λ_t held fixed, and $G^f_{t,k}$ likewise on the feature-reuse branch. Then the FAMA *map*

$$\Phi(x_k) := \alpha \sum_{t=1}^{T} \big(G_{t,k} - G^f_{t,k}\big) \quad (7)$$

equals $\partial\Delta M/\partial m_k$ at $m = \mathbf{1}$ exactly.

A positive entry of $\Phi(x_k)$ marks a region whose features raise ΔM, and thus help backbone adaptation, while a negative entry marks a harmful region. To first order, suppressing a region changes ΔM by minus the sum of Φ over it, which is the quantity that the deletion test of Section 4 probes. The local terms require the support loss as a function of both the parameters and the mask. Detaching A_k from the graph, as CAM implementations commonly do, silently zeroes the backbone block of $\nabla_\phi \mathcal{L}$ and drops the backbone part of λ_t from the attribution (Remark A.4).

Mathematically, Lemma 3.3 and Proposition 3.4 are reverse-mode differentiation through the unrolled inner loop. The contribution lies in the differentiated quantity, the mask derivative of a contrast between two branches, computed with one double-backward graph in memory at a time (Section 3.5). To first order in $T\alpha$, the map also takes a closed form at $m = \mathbf{1}$ (Appendix A.3),

$$\Phi(x_k) = T\alpha\, \partial_{m_k} \big\langle g_b, \nabla_{\phi^b} \mathcal{L}(\theta_0, S_i; m)\big\rangle + o(T\alpha). \quad (8)$$

FAMA thus localizes the support regions behind the gradient alignment or conflict of (4).

Corollary 3.5 (Completeness). *Let $\Phi^{\text{int}}(x_k) := \int_0^1 \partial\Delta M/\partial m_k|_{m=\beta\mathbf{1}}\, \mathrm{d}\beta$. Then $\sum_k \sum_{\omega\in\Omega} \Phi^{\text{int}}(x_k)_\omega = \Delta M$.*

With every mask at zero, the backbone receives no gradient and both branches coincide, so $\Delta M(\mathbf{0}) = 0$. The identity then follows from the fundamental theorem of calculus (Appendix A.3). The integrated map therefore decomposes the adaptation gain exactly across support regions. A trapezoidal rule with n_β intervals computes it at $n_\beta + 1$ times the cost of (7). Proposition 3.4 also holds for masks at earlier blocks, which give finer maps, but completeness needs the backbone output, because an earlier mask leaves the layers above it free to adapt. For display, we upsample $\Phi(x_k)$ bilinearly, keep its sign, and use one symmetric color scale per task, so that magnitudes are comparable across support images.

3.4. A unifying view

Proposition 3.4 is one instance of a general template. Let ψ be any perturbation of the support set that enters the inner-loop objective, with $\psi = 0$ the unperturbed set, and let $\Lambda = (\Lambda_t)_{t=1}^T$ be any estimate of the adjoint. We define

$$\mathcal{I}_\psi[\Lambda] := -\alpha \sum_{t=1}^{T} \Lambda_t^\top\, \partial_\psi \nabla_\phi \mathcal{L}(\phi_{t-1}, S_i^\psi)\big|_{\psi=0}. \quad (9)$$

Proposition 3.6 (Unification). *(i) With the exact adjoint, $\mathcal{I}_\psi[\lambda] = \mathrm{d}R(\phi_T)/\mathrm{d}\psi$. (ii) For example weights w, with $\mathcal{L}(\phi, S_i^w) = \frac{1}{|S_i|}\sum_k (1+w_k)\,\ell(\phi; x_k, y_k)$, $-\mathcal{I}_{w_k}[\lambda]$ is the estimator of Hara et al. (2019) for the linear influence of removing x_k from the full-batch inner loop. The myopic adjoint $\Lambda_t = \nabla R(\phi_{t-1})$ gives $\mathcal{I}_w = -\mathrm{TracIn}/|S_i|$ over the inner-loop checkpoints (Pruthi et al., 2020). (iii) If $\phi_t \to \phi^\star$, a local minimum of $\mathcal{L}(\cdot, S_i)$ with Hessian $H^\star \succ 0$ and $0 < \alpha < 2/\lambda_{\max}(H^\star)$, then as $T \to \infty$, $\mathcal{I}_\psi[\lambda] \to -\nabla R(\phi^\star)^\top (H^\star)^{-1} \partial_\psi \nabla_\phi \mathcal{L}(\phi^\star, S_i^\psi)$, the influence-function response (Koh & Liang, 2017). (iv) For the masks, $\psi = m - \mathbf{1}$, and $\mathcal{I}_m[\lambda^f]$ on the feature-reuse branch minus $\mathcal{I}_m[\lambda]$ on the full branch is the* FAMA *map* (7).

These instances differ only in the perturbation ψ and the adjoint estimate Λ. The myopic adjoint judges each update by its immediate effect on the query risk, as TracIn does, and ignores how later steps amplify or cancel it. The implicit adjoint replaces the sum by the limit in (iii), and thus assumes a converged inner loop, which fast adaptation by design is not. Applied to the masks with the feature-reuse contrast of (7), the two estimates yield the TracIn-CAM and IF-CAM baselines of Section 5, defined in Appendix C.2. Keeping the exact adjoint but perturbing the support pixels instead of the masks yields a third baseline, the pixel gradient of ΔM through the trajectory. The next result bounds when the exact adjoint matters (proofs in Appendix A.4).

Proposition 3.7 (Myopic error). *Suppose that along the trajectory $\|H_t\| \le \kappa$, $\|\nabla_\phi \mathcal{L}(\phi_t, S_i)\| \le \gamma$, $\|\nabla R(\phi_T)\| \le \gamma_R$, ∇R is κ_R-Lipschitz, $\|\partial_{\psi_j} \nabla_\phi \mathcal{L}\| \le B$, and $T\alpha\kappa \le 1$. Then, for every component ψ_j,*

$$\left|\mathcal{I}_{\psi_j}[\lambda] - \mathcal{I}_{\psi_j}[\Lambda^{\mathrm{myo}}]\right| \le \tfrac{1}{2} B\,(2\kappa\gamma_R + \kappa_R\gamma)\,T(T+1)\,\alpha^2, \tag{10}$$

whereas $|\mathcal{I}_{\psi_j}[\lambda]| \le e\,B\gamma_R\,T\alpha$.

The myopic and exact attributions thus agree to first order in $T\alpha$. Relative to the bound on $|\mathcal{I}_{\psi_j}[\lambda]|$, however, their gap is of order $T\alpha(\kappa + \kappa_R\gamma/\gamma_R)$, a product of step length and curvature. A small displacement $T\alpha\gamma$ under Assumption 3.1 therefore does not make the two adjoints interchangeable. They can diverge as the trajectory lengthens or the support loss curves more sharply, which Section 5 tests by sweeping T and α. As the next section shows, exactness comes at no extra cost.

3.5. Algorithm and cost

Algorithm 1 summarizes FAMA. The adjoint step (5) and the local term (6) are derivatives of the same scalar, $\lambda_t^\top \nabla_\phi \mathcal{L}(\phi_{t-1}, S_i; m)$, with respect to ϕ and to m, so one double-backward pass per step yields both. Per branch, FAMA costs T inner-loop steps, one query gradient, and T double-backward passes. It stores the T states, or recomputes them with checkpointing. The myopic baseline needs the same passes plus a query gradient at every step, so the exact adjoint is free in comparison. In the feature-reuse branch, all derivatives are taken with respect to the head parameters.

Algorithm 1 FAMA for one task

Require: initialization θ_0, task (S_i, Q_i), steps T, rate α
Ensure: adaptation gain ΔM, maps $\{\Phi(x_k)\}_k$

```
Roll out {φ_t} by (1) and the head-only {φ_t^f}, storing the states
ΔM ← paired bootstrap estimate of (2)
Φ ← 0
for (φ, σ) ∈ {(φ, +1), (φ^f, −1)} do
    λ ← ∇_φ R(φ_T)
    for t = T, T − 1, . . . , 1 do
        s ← λ^⊤ ∇_φ L(φ_{t−1}, S_i; m)|_{m=1}  {graph kept}
        (v, G) ← ∇_{(φ,m)} s  {one double backward}
        Φ ← Φ + σαG;    λ ← λ − αv
    end for
end for
return ΔM and Φ, upsampled to the images
```

4. Faithfulness and Ground-Truth Protocols

An attribution map is useful only if it is faithful to the mechanism it explains. Faithfulness is usually tested on a single static prediction (Samek et al., 2017; Petsiuk et al., 2018; Adebayo et al., 2018). FAMA instead explains a signed scalar produced by a trajectory, so we adapt these tests and add two that exploit cheap re-adaptation. None of the protocols is specific to FAMA, and all of them apply to any explainer of fast adaptation.

4.1. Bidirectional deletion test

We partition each support image into n_{sp} SLIC superpixels (Achanta et al., 2012) and score each superpixel by its mean attribution. Given a deletion ratio $\rho \in [0, 1]$ and an order $\pi \in \{\mathrm{desc}, \mathrm{asc}, \mathrm{rand}\}$, we replace the first $\lfloor \rho\, n_{\mathrm{sp}} \rfloor$ superpixels by a blurred copy, re-run both inner loops, and record $\Delta M(\rho, \pi)$. The areas between curves

$$\begin{aligned} \mathrm{ABC}_\downarrow &:= \textstyle\int_0^1 \left[\Delta M(\rho, \mathrm{rand}) - \Delta M(\rho, \mathrm{desc})\right] \mathrm{d}\rho, \\ \mathrm{ABC}_\uparrow &:= \textstyle\int_0^1 \left[\Delta M(\rho, \mathrm{asc}) - \Delta M(\rho, \mathrm{rand})\right] \mathrm{d}\rho \end{aligned} \tag{11}$$

measure whether deleting the most helpful regions lowers ΔM, and deleting the most harmful ones raises it, faster than chance. A signed map makes both claims, so BI-ADT requires both areas to be positive. Deletion acts on pixels, which also tests how well a feature-level map transfers to input interventions. We compare explainers by paired differences of the two areas on the same tasks.

4.2. Counterfactual re-adaptation

An inner loop of T steps is cheap enough to re-run after every edit of the support set, which yields the true effect of the edit on ΔM. At the image level, we replace x_k by its blurred copy and record the resulting gain ΔM_{-k}. We replace rather than remove, because removing the only image of a class in the 1-shot setting would remove the class. At the region level, we blur one superpixel at a time on a subset of tasks. The score is the Spearman correlation between the attribution mass of an image or region and its true effect $\Delta M - \Delta M_{-k}$. Each edit costs $2T$ inner-loop steps, and example-level methods (TracIn, influence functions, and SGD-influence) are scored on images only.

4.3. Planted-feature benchmark

We plant a synthetic patch at a random position in the support images of one class and record its mask (Adebayo et al., 2022). A *helpful* patch also appears on the query images of that class, so adapting to it lowers the query risk. A *harmful* patch appears only in the support set, or on query images of another class, so adapting to it raises the risk. We keep only tasks in which the patch moves ΔM in the designed direction. Localization is scored by the AUROC of the attribution against the patch mask, and the sign by the fraction of patches whose total attribution has the designed sign. Patch size, contrast, position, and the fraction of affected images set the difficulty, and off-center positions rule out a center prior.

4.4. Sanity checks

Cascading randomization of the backbone, from the output toward the input, must decorrelate the maps (Adebayo et al., 2018). We also randomize each layer independently and meta-train with permuted labels, the data-randomization test of the same work. Finally, we edit the support set by label permutation, blur, and cross-domain mixing. Label permutation mimics the label-flipped noisy tasks of Yao et al. (2021). We report the change of the map together with the paired change of ΔM.

5. Experiments

We ask three questions. Is FAMA more faithful than static and example-level explainers (Section 5.2)? When does the exact adjoint matter (Section 5.3)? Does contextual bias cause negative transfer, and do edits guided by FAMA mitigate it (Section 5.4)? Finally, we apply FAMA beyond MAML, to LoRA adaptation of CLIP on dermoscopy images (Section 5.5).

5.1. Setup

Models. We meta-train MAML (Finn et al., 2017) on 5-way 1-shot tasks with a Conv4 backbone under the original protocol. The protocol uses inner rate $\alpha = 0.01$, outer Adam rate 10^{-3}, 5 inner steps in training and $T = 10$ at test time, meta-batches of 4 tasks, and 60,000 iterations. A ResNet-12 backbone (Lee et al., 2019) tests robustness to depth. For few-shot LoRA adaptation (Hu et al., 2022) of CLIP ViT-B/16 (Radford et al., 2021), the feature-reuse branch trains only a linear head on frozen features.

Data. We meta-train on miniImageNet (Vinyals et al., 2016; Ravi & Larochelle, 2017) or tieredImageNet (Ren et al., 2018) and draw cross-domain test tasks from CUB-200-2011 (Wah et al., 2011), following Chen et al. (2019). Waterbirds (Sagawa et al., 2020) provides a controlled context shift. For CLIP, we use the ISIC dermoscopy tasks of the BSCD-FSL benchmark (Guo et al., 2020).

Baselines. The baselines vary the two choices of Section 3.4, namely what is perturbed and which adjoint is used. At the feature map, we compare Grad-CAM at ϕ_T (Selvaraju et al., 2017), which uses no adjoint, with TracIn-CAM and IF-CAM, which use the myopic and the implicit adjoint, respectively. At the pixels, we take the gradient of ΔM through the trajectory with integrated gradients (Sundararajan et al., 2017). At the level of examples, we compare TracIn (Pruthi et al., 2020), influence functions (Koh & Liang, 2017), and SGD-influence (Hara et al., 2019), each computed on both branches (Appendix C.2). Random maps and a center prior serve as controls. All methods reuse the same trajectories.

Protocol. Each configuration uses 600 meta-test tasks and three meta-training seeds, and ΔM is estimated as in Appendix C.1. We compare methods by paired Wilcoxon tests with Holm correction and report 95% bootstrap confidence intervals.

Negative transfer is common. Backbone adaptation hurts on 49.7% of tieredImageNet→CUB tasks and on 12.5% of miniImageNet→CUB tasks. Cross-domain transfer thus supplies many tasks of both signs to explain, so we evaluate on tieredImageNet→CUB unless stated otherwise.

5.2. Faithfulness against baselines

Table 1 compares all explainers. **[Headline: FAMA vs. the strongest baseline on each metric.]** Example-level methods rank support images but cannot localize evidence within them. **[Compare ρ_{img} of SGD-influence and FAMA, as predicted by Proposition 3.6.]** The pixel gradient of ΔM follows the trajectory exactly but is noisy. **[Compare ρ_{reg} and AUROC to FAMA.]** All spatial explainers pass

Table 1. Faithfulness on tieredImageNet→CUB (Conv4, 5-way 1-shot; 600 tasks × 3 seeds). BI-ADT areas (11), Spearman correlation with re-adaptation ground truth at image and region level (Section 4.2), and localization and sign accuracy on planted features (Section 4.3). Best in bold; * marks a significant paired difference to the runner-up after Holm correction. **[Placeholder values.]**

	Method	ABC$_\downarrow$	ABC$_\uparrow$	ρ_{img}	ρ_{reg}	AUROC	Sign acc.	Time (s)
Controls	Random	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	–
	Center prior	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	–
Example	TracIn	–	–	[ ]	–	–	[ ]	[ ]
	Influence function	–	–	[ ]	–	–	[ ]	[ ]
	SGD-influence	–	–	[ ]	–	–	[ ]	[ ]
Spatial	Grad-CAM at ϕ_T	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
	$\partial\Delta M/\partial x$ (integrated gradients)	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
	TracIn-CAM	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
	IF-CAM	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Ours	FAMA	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
	FAMA (integrated, n_β=16)	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

[Placeholder]
Region-level agreement ρ_{reg} (y) vs. $T\alpha$ (x, log scale) for Grad-CAM at ϕ_T, TracIn-CAM, IF-CAM, and FAMA; one panel per backbone.

Figure 2. Agreement with re-adaptation ground truth as the trajectory lengthens. Proposition 3.7 predicts that TracIn-CAM matches FAMA for small $T\alpha\kappa$ and departs from it as $T\alpha\kappa$ grows, where κ bounds the support-loss curvature. **[Placeholder.]**

Table 2. Left: accuracy recovered on negative-gain tasks by blurring the support regions an explainer marks as harmful, with the explainer run on a held-out half of the query set. The example-level explainer blurs whole images. Right: BI-ADT for few-shot LoRA adaptation of CLIP on ISIC. **[Placeholder values.]**

	Support editing	CLIP + LoRA, ISIC	
Explainer	Recovered (%)	ABC$_\downarrow$	ABC$_\uparrow$
Random	[ ]	[ ]	[ ]
Grad-CAM at ϕ_T	[ ]	[ ]	[ ]
TracIn-CAM	[ ]	[ ]	[ ]
SGD-influence	[ ]	–	–
FAMA	[ ]	[ ]	[ ]

the cascading-randomization check **[or not; report in Appendix C.6]**, and results on ResNet-12 follow the same ranking (Appendix C.4) **[result]**.

5.3. When does the exact adjoint matter?

We sweep $T \in \{1, 5, 10, 20\}$ and $\alpha \in \{0.005, 0.01, 0.02\}$ at test time (Figure 2). **[Result: the gap between TracIn-CAM and FAMA as a function of $T\alpha$, and the point at which end-state Grad-CAM stops tracking ΔM.]** Ablations in Appendix C.5 remove one component at a time, namely the feature-reuse contrast, the sign, the sum over steps, and the backbone block of the adjoint. **[Result: which component matters most.]**

5.4. Contextual bias as a cause of negative transfer

On CUB, positive-gain maps concentrate on the bird, whereas negative-gain maps shift onto branches, foliage, and artifacts (Figure 3a). Using the bounding boxes of CUB-200-2011, we measure the share of harmful attribution that falls on the background. **[Result: correlation of this share with ΔM over 1,200 tasks.]** To test causality, we build Waterbirds tasks whose support backgrounds correlate with the labels at strength c while the query backgrounds do not. **[Result: ΔM falls and the background share of FAMA rises with c (Figure 3c).]**

Finally, we act on the explanations. On negative-gain tasks, we blur the support regions that an explainer marks as harmful and adapt again. The explainer sees only half of the query set, and accuracy is measured on the other half, so no query label leaks into the edit. **[Result: FAMA recovers $X\%$ of the accuracy lost to negative transfer, compared with $Y\%$ for the best baseline (Table 2, left).]**

5.5. Beyond MAML: LoRA adaptation of CLIP

The template of Section 3.4 needs only a differentiable update, so FAMA applies unchanged to few-shot LoRA adaptation of CLIP (Appendix B.2). We test it on ISIC, the kind of medical task that motivates this work (Section 1). **[Result: BI-ADT of FAMA and the spatial baselines (Table 2, right), and one task in which harmful attribution falls on rulers or ink markers.]**

[Placeholder] (a) FAMA maps on the most positive- and most negative-gain tieredImageNet→CUB tasks (red: helpful, blue: harmful).
(b) Background share of harmful attribution vs. ΔM over 1,200 cross-domain tasks. (c) Waterbirds: ΔM and background share vs. support background–label correlation.

Figure 3. Contextual bias drives negative transfer. **[Placeholder caption: state the observed trend and its effect size.]**

6. Discussion and Conclusion

Limitations. ΔM measures performance, so it needs labeled evaluation data. When only the support set is labeled, a split of S_i estimates ΔM at the price of bias and variance, and it is unavailable in the 1-shot setting. A label-free proxy, such as prediction entropy, would turn FAMA into an explanation of the proxy rather than of performance. The resolution of Φ is that of the feature grid, and masks at earlier blocks give finer maps at the cost of completeness. Time grows linearly with T and memory with TP, so long inner loops call for checkpointing or the implicit adjoint. Finally, the attributions are first-order, and causal claims rest on the interventions of Section 4.

Outlook. The template of Section 3.4 applies to any differentiable adaptation with a counterfactual baseline. Explaining when test-time training beats in-context learning (Akyürek et al., 2025), or when a self-generated edit helps (Zweiger et al., 2025), is a natural next step.

Conclusion. We turned negative transfer in meta-learning from an accuracy symptom into an explanation. The adaptation gain tells whether backbone adaptation helped on a task, and FAMA tells which support regions made it so, as an exact derivative that integrates to the gain itself. Protocols with counterfactual ground truth show that **[headline result]**, and the explanations expose contextual bias and guide edits that mitigate it.

Impact Statement

This work aims to make few-shot adaptation auditable. In medical imaging, an explanation of why adaptation failed on a task could help practitioners detect spurious cues, such as acquisition artifacts, before deployment. Attribution maps can also be over-trusted. We therefore evaluate them against counterfactual ground truth and recommend that they complement, rather than replace, clinical validation.

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A. Proofs

This appendix proves the results of Section 3 in the order of the main text. Appendix A.1 states the standing assumptions and three auxiliary lemmas. Appendices A.2 to A.4 then treat the adaptation gain, the attribution map, and the unifying view.

A.1. Standing assumptions and auxiliary lemmas

Assumptions. Throughout, $\mathcal{L}(\phi, S_i; m)$ is twice continuously differentiable in (ϕ, m) on a neighborhood of the trajectories of both branches, for every mask considered, including $m = \beta\mathbf{1}$ with $\beta \in [0, 1]$. The query risk R is twice continuously differentiable on a neighborhood of θ_0 that contains these trajectories. Proposition 3.2 and (8) additionally use Assumption 3.1. With ReLU activations or max-pooling, these conditions hold on each region where the activation pattern is fixed. The derivatives in Lemma 3.3 and Propositions 3.4 and 3.6 are then exact almost everywhere, whereas Proposition 3.2 requires smooth activations or trajectories that stay within one region.

Notation. We write $J_s := I - \alpha H_s$ for the step Jacobian and $g = (g_b, g_h) := \nabla R(\theta_0)$ for the blocks of the risk gradient. The displacements after t steps are $\Delta_t^b := \phi_t^b - \theta_0^b$, $\Delta_t^h := \phi_t^h - \theta_0^h$, and $\Delta_t^{fh} := \phi_t^{f,h} - \theta_0^h$, with $\Delta^b = \Delta_T^b$, $\Delta^h = \Delta_T^h$, and $\Delta^{fh} = \Delta_T^{fh}$. The head divergence is $\delta_t := \Delta_t^h - \Delta_t^{fh}$. Products of Jacobians are ordered as $J_t \cdots J_{T-1}$, and an empty product is the identity.

Lemma A.1 (Displacement). *If $\|\nabla_\phi \mathcal{L}(\cdot, S_i)\| \le \gamma$ along both trajectories, then $\|\phi_s - \phi_t\| \le |s - t|\,\alpha\gamma$ and $\|\phi_s^f - \phi_t^f\| \le |s - t|\,\alpha\gamma$ for all s, t. In particular, $\|\Delta_t^b\|$, $\|\Delta_t^h\|$, and $\|\Delta_t^{fh}\|$ are at most $t\alpha\gamma$.*

Proof. Each step of (1), and each head-only step, moves the parameters by α times a gradient of norm at most γ. The triangle inequality over $|s - t|$ steps gives the first two bounds. The last one follows with $s = 0$, since a block has at most the norm of the full vector. □

Lemma A.2 (Head divergence). *If $\nabla_{\phi^h} \mathcal{L}(\cdot, S_i)$ is κ-Lipschitz and Lemma A.1 applies, then $\delta_1 = 0$ and $\|\delta_T\| \le \frac{1}{2} e^{T\alpha\kappa} \kappa\gamma\, T^2\alpha^2$.*

Proof. Both branches start at θ_0, so their first head steps coincide and $\delta_1 = 0$. For $t \ge 2$, $\delta_t = \delta_{t-1} - \alpha\big[\nabla_{\phi^h} \mathcal{L}(\phi_{t-1}, S_i) - \nabla_{\phi^h} \mathcal{L}(\phi_{t-1}^f, S_i)\big]$. The two arguments differ by Δ_{t-1}^b in the backbone and by δ_{t-1} in the head, so Lipschitzness and Lemma A.1 give $\|\delta_t\| \le (1 + \alpha\kappa)\|\delta_{t-1}\| + \alpha\kappa(t - 1)\alpha\gamma$. Unrolling from $\delta_1 = 0$ yields $\|\delta_T\| \le \kappa\gamma\alpha^2 \sum_{t=2}^{T} (t - 1)(1 + \alpha\kappa)^{T-t} \le \frac{1}{2} e^{T\alpha\kappa} \kappa\gamma\, T^2\alpha^2$. □

Lemma A.3 (Jacobian products). *If $\|H_s\| \le \kappa$ for all s and $n\alpha\kappa \le 1$, then $\|J_t \cdots J_{t+n-1} - I\| \le e^{n\alpha\kappa} - 1 \le 2n\alpha\kappa$. The same holds for the head blocks of the Hessians.*

Proof. Write $J_s = I + E_s$ with $\|E_s\| \le \alpha\kappa$. Expanding the product into a sum over subsets of factors and removing the empty subset, which contributes I, the triangle inequality gives $\|\prod_s (I + E_s) - I\| \le \prod_s (1 + \|E_s\|) - 1 \le (1 + \alpha\kappa)^n - 1 \le e^{n\alpha\kappa} - 1$. By convexity, $e^x - 1 \le (e - 1)x \le 2x$ for $x \in [0, 1]$. A head block has at most the norm of the full Hessian. □

A.2. Proof of Proposition 3.2

Proof. By Assumption 3.1, Taylor's theorem expands $R(\phi_T)$ and $R(\phi_T^f)$ around θ_0 to second order, each with a remainder bounded by $\kappa_H \|\phi - \theta_0\|^3/6$. The feature-reuse branch has no backbone displacement. Subtracting the two expansions cancels $R(\theta_0)$ and gives

$$\Delta M = -C_b + \underbrace{g_h^\top (\Delta^{fh} - \Delta^h) + \tfrac{1}{2}\big(\Delta^{fh\top} H_{hh}^R \Delta^{fh} - \Delta^{h\top} H_{hh}^R \Delta^h\big)}_{R_h} + \varepsilon_3, \tag{12}$$

where ε_3 is the difference of the two remainders. By Lemma A.1, $C_b = O(T\alpha)$ and $|\varepsilon_3| \le \frac{1}{3}\kappa_H (T\alpha\gamma)^3$. The linear part of R_h equals $-g_h^\top \delta_T$, which is $O((T\alpha)^2)$ by Lemma A.2. The quadratic part equals $-\frac{1}{2}\delta_T^\top H_{hh}^R (\Delta^{fh} + \Delta^h) = O((T\alpha)^3)$. This proves (3). For $T = 1$, $\delta_1 = 0$ gives $R_h = 0$ exactly, so the remainder reduces to $\varepsilon_3 = O(\alpha^3)$.

For (4), write $\Delta^b = -\alpha \sum_{t=0}^{T-1} \nabla_{\phi^b} \mathcal{L}(\phi_t, S_i)$. By Lemma A.1, $\|\phi_t - \theta_0\| \le t\alpha\gamma$, so the κ-Lipschitz support gradient gives $\Delta^b = -T\alpha \nabla_{\phi^b} \mathcal{L}(\theta_0, S_i) + e$ with $\|e\| \le \frac{1}{2}\kappa\gamma T^2\alpha^2$. The two quadratic terms of C_b are $O((T\alpha)^2)$ by Lemma A.1. Hence $-C_b = -g_b^\top \Delta^b + O((T\alpha)^2) = T\alpha \langle g_b, \nabla_{\phi^b} \mathcal{L}(\theta_0, S_i)\rangle + O((T\alpha)^2)$. □

A.3. Proofs for the attribution map

This subsection proves Lemma 3.3, Proposition 3.4, the first-order form (8), and Corollary 3.5.

Proof of Lemma 3.3. By backward induction on t, starting from $\lambda_T = \partial R(\phi_T)/\partial \phi_T$. Since ϕ_T depends on ϕ_{t-1} only through ϕ_t, the chain rule gives $\partial R(\phi_T)/\partial \phi_{t-1} = J_{t-1}^\top \lambda_t = \lambda_t - \alpha H_{t-1}\lambda_t$, because H_{t-1} is symmetric. No approximation enters, since (1) is a literal map rather than a discretized ODE. □

Proof of Proposition 3.4. The masks enter the trajectory only through the update maps $U_t(\phi, m) := \phi - \alpha \nabla_\phi \mathcal{L}(\phi, S_i; m)$, since $\phi_0 = \theta_0$ does not depend on m and the query risk is evaluated without masks. The total derivative therefore obeys $\mathrm{d}\phi_t/\mathrm{d}m_k = J_{t-1}\,\mathrm{d}\phi_{t-1}/\mathrm{d}m_k + \partial U_t/\partial m_k$ with $\mathrm{d}\phi_0/\mathrm{d}m_k = 0$. Unrolling gives $\mathrm{d}\phi_T/\mathrm{d}m_k = \sum_{t=1}^T (\partial \phi_T/\partial \phi_t)\,\partial U_t/\partial m_k$, and Lemma 3.3 turns this into

$$\frac{\mathrm{d}R(\phi_T)}{\mathrm{d}m_k} = \sum_{t=1}^T \lambda_t^\top \frac{\partial U_t}{\partial m_k} = -\alpha \sum_{t=1}^T \frac{\partial}{\partial m_k}\Big[\lambda_t^\top \nabla_\phi \mathcal{L}(\phi_{t-1}, S_i; m)\Big] = -\alpha \sum_{t=1}^T G_{t,k},$$

where λ_t is held fixed inside the bracket. The same argument on the feature-reuse branch, whose state is the head alone, gives $\mathrm{d}R(\phi_T^f)/\mathrm{d}m_k = -\alpha \sum_t G_{t,k}^f$. Subtracting the first from the second according to (2) yields $\partial \Delta M/\partial m_k = \alpha \sum_t (G_{t,k} - G_{t,k}^f) = \Phi(x_k)$. □

Remark A.4 (Detaching the feature map). With $z_k = m_k \odot A_k$ and $A_k = f_{\phi^b}(x_k)$, the backbone gradient of the support loss is $\nabla_{\phi^b}\mathcal{L} = \sum_k (\partial A_k/\partial \phi^b)^\top (m_k \odot \partial \mathcal{L}/\partial z_k)$. Detaching A_k sets $\partial A_k/\partial \phi^b = 0$, so this gradient vanishes. The mask then loses its backbone contributions to $G_{t,k}$, namely the term $\langle (\partial A_k/\partial \phi^b)\lambda_t^b, \partial \mathcal{L}/\partial z_k\rangle$ at each location and the curvature term through $\partial^2 \mathcal{L}/\partial z_k^2$.

Proof of (8). By Lemma A.1, every state of both branches lies within $T\alpha\gamma$ of θ_0. The adjoint is $\lambda_t = J_t \cdots J_{T-1} \nabla R(\phi_T)$, so Lemma A.3 and the continuity of ∇R give $\lambda_t \to g$ as $T\alpha \to 0$, uniformly in t. Likewise, $\lambda_t^f \to g_h$. Since $\partial_m \nabla_\phi \mathcal{L}$ is continuous, the local terms satisfy

$$G_{t,k} - G_{t,k}^f = \frac{\partial}{\partial m_k}\Big[g^\top \nabla_\phi \mathcal{L}(\theta_0, S_i; m) - g_h^\top \nabla_{\phi^h} \mathcal{L}(\theta_0, S_i; m)\Big]_{m=\mathbf{1}} + o(1)$$

uniformly in t. The bracket equals $\langle g_b, \nabla_{\phi^b}\mathcal{L}(\theta_0, S_i; m)\rangle$, and summing α times the local terms over the T steps gives (8). □

Proof of Corollary 3.5. Let $F(\beta) := \Delta M(\beta \mathbf{1})$, which is continuously differentiable on $[0, 1]$. Proposition 3.4 holds verbatim at any m, with the trajectories and adjoints of the rollout under that mask, so the integrand is computable. By the chain rule, $F'(\beta) = \sum_k \sum_\omega \partial \Delta M/\partial m_{k,\omega}|_{m=\beta\mathbf{1}}$, and the fundamental theorem of calculus gives $\sum_k \sum_\omega \Phi^{\mathrm{int}}(x_k)_\omega = F(1) - F(0)$. At $m = \mathbf{0}$, every z_k vanishes, so $\nabla_{\phi^b}\mathcal{L} = 0$ and $\phi_t^b = \theta_0^b$ for all t. Both branches then apply identical head updates from identical states, so $\phi_t = \phi_t^f$ for all t and $F(0) = 0$. □

A.4. Proofs for the unifying view

Proof of Proposition 3.6. (i) The proof of Proposition 3.4 uses only that the perturbation enters through the update maps. It therefore applies verbatim to any ψ and gives $\mathrm{d}R(\phi_T)/\mathrm{d}\psi = \mathcal{I}_\psi[\lambda]$.

(ii) For example weights, $\partial_{w_k} \nabla_\phi \mathcal{L}(\phi, S_i^w) = \nabla_\phi \ell(\phi; x_k, y_k)/|S_i|$. Hence $\mathcal{I}_{w_k}[\lambda] = -\frac{\alpha}{|S_i|}\sum_t \lambda_t^\top \nabla_\phi \ell(\phi_{t-1}; x_k, y_k)$. Hara et al. (2019) estimate the effect of removing an example by propagating a backward vector through the steps of SGD. Their counterfactual drops the gradient of x_k but keeps the normalization by the batch size, which corresponds to $w_k = -1$. Their first-order estimator, in its multi-epoch form and specialized to full-batch steps with rate α, is therefore $\frac{\alpha}{|S_i|}\sum_t \lambda_t^\top \nabla_\phi \ell(\phi_{t-1}; x_k, y_k) = -\mathcal{I}_{w_k}[\lambda]$. With $\Lambda_t = \nabla R(\phi_{t-1})$, $\mathcal{I}_{w_k} = -\frac{1}{|S_i|}\sum_t \alpha\, \nabla R(\phi_{t-1})^\top \nabla_\phi \ell(\phi_{t-1}; x_k, y_k)$. The sum is TracIn with checkpoints $\phi_0, \ldots, \phi_{T-1}$ and step size α (Pruthi et al., 2020).

(iii) The adjoint is $\lambda_t = J_t J_{t+1} \cdots J_{T-1} \nabla R(\phi_T)$. Since $\phi_s \to \phi^\star$ and $0 < \alpha < 2/\lambda_{\max}(H^\star)$ with $H^\star \succ 0$, there are s_0 and $r < 1$ with $\|J_s\| \le r$ for all $s \ge s_0$. The terms with $t < s_0$ are bounded by a constant times r^{T-s_0} and vanish as

$T \to \infty$. For the remaining terms, $J_s \to J^\star := I - \alpha H^\star$, $\nabla R(\phi_T) \to \nabla R(\phi^\star)$, and $\partial_\psi \nabla_\phi \mathcal{L}(\phi_{t-1}) \to \partial_\psi \nabla_\phi \mathcal{L}(\phi^\star)$, and the geometric decay makes the convergence uniform in t. The sum therefore tends to

$$-\alpha \sum_{j=0}^{\infty} \nabla R(\phi^\star)^\top (J^\star)^j \, \partial_\psi \nabla_\phi \mathcal{L}(\phi^\star) = -\nabla R(\phi^\star)^\top (H^\star)^{-1} \partial_\psi \nabla_\phi \mathcal{L}(\phi^\star),$$

by the Neumann series $\alpha \sum_{j \geq 0} (I - \alpha H^\star)^j = (H^\star)^{-1}$. This is the influence-function response of Koh & Liang (2017).

(iv) With $\psi = m - \mathbf{1}$, $\partial_\psi = \partial_m$, and $\psi = 0$ corresponds to $m = \mathbf{1}$. Hence $\mathcal{I}_{m_k}[\lambda] = -\alpha \sum_t G_{t,k}$ by (6), and $\mathcal{I}_{m_k}[\lambda^f] = -\alpha \sum_t G^f_{t,k}$ on the feature-reuse branch. The second minus the first is $\alpha \sum_t (G_{t,k} - G^f_{t,k}) = \Phi(x_k)$, in agreement with Proposition 3.4. □

Proof of Proposition 3.7. Fix t and let $n = T - t$. The exact adjoint is $\lambda_t = J_t \cdots J_{T-1} \nabla R(\phi_T)$, so Lemma A.3 gives $\|\lambda_t - \nabla R(\phi_T)\| \leq 2n\alpha\kappa\gamma_R$. By Lemma A.1, $\|\phi_T - \phi_{t-1}\| \leq (n+1)\alpha\gamma$, so $\|\nabla R(\phi_T) - \nabla R(\phi_{t-1})\| \leq \kappa_R (n+1)\alpha\gamma$. The triangle inequality gives

$$\|\lambda_t - \Lambda_t^{\text{myo}}\| \leq (2\kappa\gamma_R + \kappa_R\gamma)(T - t + 1)\,\alpha.$$

Summing over t with $\|\partial_{\psi_j} \nabla_\phi \mathcal{L}\| \leq B$ yields $|\mathcal{I}_{\psi_j}[\lambda] - \mathcal{I}_{\psi_j}[\Lambda^{\text{myo}}]| \leq \alpha B(2\kappa\gamma_R + \kappa_R\gamma)\,\alpha \sum_{t=1}^{T}(T - t + 1) = \frac{1}{2} B(2\kappa\gamma_R + \kappa_R\gamma)\,T(T+1)\,\alpha^2$. Finally, $\|\lambda_t\| \leq (1 + \alpha\kappa)^T \gamma_R \leq e\,\gamma_R$, so $|\mathcal{I}_{\psi_j}[\lambda]| \leq e\,B\gamma_R\,T\alpha$. □

B. Further Analysis of the Adjoint

This appendix extends Sections 3.2 and 3.3 in three directions. Appendix B.1 bounds how errors propagate through the adjoint, Appendix B.2 derives the adjoint for other update rules, and Appendix B.3 delimits the tasks and architectures that the results cover.

B.1. Numerical stability

Lemma 3.3 is exact in exact arithmetic. In practice, λ_T is estimated from a finite query set, and every Hessian-vector product carries rounding error. Let $c_t := \|I - \alpha H_t\|$.

Proposition B.1 (Error propagation). *Let $\hat{\lambda}_T = \lambda_T + e_T$, and let each computed Hessian-vector product carry an error of norm at most ε_t. Then $e_t := \hat{\lambda}_t - \lambda_t$ satisfies*

$$\|e_t\| \leq \Big(\prod_{r=t}^{T-1} c_r\Big)\|e_T\| + \alpha \sum_{s=t+1}^{T} \varepsilon_s \prod_{r=t}^{s-2} c_r, \qquad t = 0, \ldots, T, \tag{13}$$

with an empty product equal to one. The map of one branch deviates by at most $\alpha\, B_m \sum_{t=1}^{T} \|e_t\|$, where B_m bounds $\|\partial_m \nabla_\phi \mathcal{L}\|$.

Proof. The computed recursion is $\hat{\lambda}_{t-1} = \hat{\lambda}_t - \alpha(H_{t-1}\hat{\lambda}_t + \eta_t)$ with $\|\eta_t\| \leq \varepsilon_t$. Subtracting (5) gives $e_{t-1} = (I - \alpha H_{t-1})e_t - \alpha\eta_t$, so $\|e_{t-1}\| \leq c_{t-1}\|e_t\| + \alpha\varepsilon_t$, and unrolling gives the bound. The local terms are linear in λ_t, so the error of the map is $\alpha \sum_t \partial_m [e_t^\top \nabla_\phi \mathcal{L}]$, whose norm is at most $\alpha B_m \sum_t \|e_t\|$. □

The same argument without HVP errors gives $\|\lambda_t\| \leq (\prod_{r=t}^{T-1} c_r)\|\lambda_T\|$, so the product of the c_r is the amplification factor of the recursion. If $H_r \succeq 0$ and $\alpha \leq 2/\lambda_{\max}(H_r)$, the eigenvalues of $I - \alpha H_r$ lie in $[-1, 1]$, and the step is non-expansive. Along a direction of negative curvature, however, the step expands the adjoint. Under Assumption 3.1, the product is at most $(1 + \alpha\kappa)^T \leq e$, but it need not be below one, since the support-loss Hessian is often indefinite. The query-sampling error $\|e_T\|$ shrinks as $|Q_i|^{-1/2}$, and we run the checks of Appendix C.3 in double precision.

B.2. General update rules

Let the inner loop update an augmented state s_t, such as the parameters together with optimizer moments, by a differentiable map $s_t = U_t(s_{t-1}, m)$ from a mask-independent s_0. The proof of Proposition 3.4 applies verbatim. With $\bar{\lambda}_T := \partial R / \partial s_T$ and $\bar{\lambda}_{t-1} := (\partial U_t / \partial s_{t-1})^\top \bar{\lambda}_t$, the exact attribution is $\mathrm{d}R/\mathrm{d}m_k = \sum_t (\partial U_t / \partial m_k)^\top \bar{\lambda}_t$. Each step costs one vector-Jacobian product of U_t, which automatic differentiation provides. We list common cases.

Per-parameter and per-step learning rates. For the rates α_t of Meta-SGD (Li et al., 2017) and MAML++ (Antoniou et al., 2019), the adjoint is the one given in Section 3.2. The local term of (7) becomes $\partial_{m_k}[(\alpha_t \odot \lambda_t)^\top \nabla_\phi \mathcal{L}]$, which already contains the rate.

Momentum. With momentum coefficient μ, $v_0 = 0$, $v_t = \mu v_{t-1} + \nabla_\phi \mathcal{L}(\phi_{t-1}, S_i; m)$, and $\phi_t = \phi_{t-1} - \alpha v_t$, let $\lambda_t := \partial R/\partial \phi_t$ and $\nu_t := \partial R/\partial v_t$, so that $\lambda_T = \nabla R(\phi_T)$ and $\nu_T = 0$. Differentiating both updates and writing $\xi_t := \alpha\lambda_t - \nu_t$ gives

$$\lambda_{t-1} = \lambda_t - H_{t-1}\,\xi_t, \qquad \nu_{t-1} = -\mu\,\xi_t, \qquad \frac{\mathrm{d}R}{\mathrm{d}m_k} = -\sum_{t=1}^{T} \frac{\partial}{\partial m_k}\Big[\xi_t^\top \nabla_\phi \mathcal{L}(\phi_{t-1}, S_i; m)\Big]. \tag{14}$$

For $\mu = 0$, $\nu_t = 0$ and $\xi_t = \alpha\lambda_t$, which recovers (5) and Proposition 3.4.

Adam. The normalization by the second moment makes the update non-linear in the gradient. The augmented state then holds the parameters and both moments, and the step Jacobian is obtained by automatic differentiation of U_t, at the cost of one additional elementwise vector-Jacobian product per step.

Low-rank adapters. With LoRA (Hu et al., 2022), the stored states and adjoints scale with the number of adapter parameters rather than with P. Each Hessian-vector product, however, still requires a double-backward pass through the whole network.

B.3. Scope

Lemma 3.3 and Proposition 3.4 use no property of classification. They need only a differentiable inner-loop update, a support loss that is differentiable in the mask and twice so in the parameters, a differentiable query risk, and a split into backbone and head. Regression, segmentation, and detection losses qualify, and a non-differentiable metric is replaced by a surrogate (Section 3.1). The grid Ω need not be spatial and can index tokens, time steps, or graph nodes. For a ViT, it is the patch grid. For models without a canonical head, the feature-reuse branch is a modeling choice, and each choice makes ΔM answer a different question. FAMA does not apply to learners that adapt without gradient steps, such as prototype classifiers (Snell et al., 2017) or in-context learning (Brown et al., 2020).

When completeness fails. Corollary 3.5 needs $\Delta M(\mathbf{0}) = 0$, which fails for masks at earlier blocks (Section 3.3) and in two further cases. A ViT head that reads only the class token places the patch grid inside the backbone, whereas a head on mean-pooled patch tokens restores the corollary. A head may also read features that bypass the mask, such as the skip connections of a U-Net (Ronneberger et al., 2015) or the multi-scale features of a feature pyramid (Lin et al., 2017). In these cases, Proposition 3.4 still holds, and completeness holds relative to $\Delta M(\mathbf{0})$.

C. Experimental Details and Additional Results

This appendix supplements Section 5. Appendices C.1 and C.2 give the setup and the baselines, Appendix C.3 verifies the results of Section 3 numerically, and Appendices C.4 to C.6 report further results.

C.1. Setup

Adaptation gain. Each task provides a pool of labeled query images with an equal number per class. We draw n_{boot} resamples of the pool, stratified by class, and evaluate both branches on the same resample. The estimate is the mean of $R_j^f - R_j$ over resamples $j = 1, \ldots, n_{\mathrm{boot}}$, with a percentile confidence interval. For comparability across tasks, we also report $\Delta M\% := 100\,\Delta M/R(\phi_T)$. For a non-differentiable metric, ΔM is reported with the metric itself, while the attribution uses the surrogate. The adjoint seed λ_T is the gradient of the query risk on the full pool. Gradients are linear in the loss, so resampling cannot sharpen it, and we use bootstrap gradients only to estimate the uncertainty of the maps.

Implementation. For Conv4, the mask multiplies the last convolutional feature map. For ResNet-12, it multiplies the output of the last residual block, before global average pooling. For CLIP, **[to fill: the LoRA rank and adapted layers, the number of shots, the inner-loop schedule, and the token grid that the mask multiplies]**. For MAML, both inner loops are rolled out with first-order gradients, and the states are stored as detached leaf tensors. BatchNorm stays in the same

mode during rollout, adjoint, and query evaluation. MAML uses batch statistics throughout, and automatic differentiation then includes the coupling between support images through those statistics.

Protocols. **[To fill: the number and compactness of SLIC superpixels; the blur kernel; the grid of deletion ratios ρ and the quadrature of (11); the task subset for region-level re-adaptation; the size, contrast, position, and affected fraction of the planted patches; and the randomization schedule of the sanity checks.]**

C.2. Baselines

Template baselines. TracIn-CAM and IF-CAM keep the masks and the feature-reuse contrast of (7) and change only the adjoint. TracIn-CAM replaces λ_t in the local terms (6) by the myopic adjoint $\nabla R(\phi_{t-1})$, and λ_t^f by $\nabla_{\phi^h} R(\phi_{t-1}^f)$. IF-CAM evaluates the limit of Proposition 3.6(iii) at the endpoint of each branch,

$$\Phi^{\mathrm{IF}}(x_k) := \frac{\partial}{\partial m_k}\Big[\nabla R(\phi_T)^\top (H_T + \tau I)^{-1} \nabla_\phi \mathcal{L}(\phi_T, S_i; m) - \nabla_{\phi^h} R(\phi_T^f)^\top (H_T^f + \tau I)^{-1} \nabla_{\phi^h} \mathcal{L}(\phi_T^f, S_i; m)\Big]_{m=\mathbf{1}}, \quad (15)$$

where H_T^f is the head block of the Hessian at ϕ_T^f, and both inverse-Hessian factors are held fixed. The damping $\tau > 0$ is needed, as in Koh & Liang (2017), because H_T need not be positive definite after a few steps. Each inverse-Hessian-vector product is computed iteratively from Hessian-vector products.

Example-level baselines. TracIn, influence functions, and SGD-influence score the effect of an example on the risk of one branch (Proposition 3.6(ii)), whereas the re-adaptation ground truth $\Delta M - \Delta M_{-k}$ of Section 4.2 contrasts two branches. We therefore compute each score on both branches and take the feature-reuse score minus the full-branch score, as in (7). On the feature-reuse branch, all gradients are taken with respect to the head parameters, and influence functions use the damped inverse Hessians of (15).

C.3. Numerical verification

Exactness of Proposition 3.4. For each task, we draw a random direction u in mask space with $\|u\| = 1$ and compare $\langle \Phi, u \rangle$ with the central difference $[\Delta M(\mathbf{1} + \varepsilon u) - \Delta M(\mathbf{1} - \varepsilon u)]/(2\varepsilon)$ for $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$. **[Result: median and maximum relative error over N tasks, for the exact map and for the variant that detaches A_k.]**

Completeness. **[Result: relative residual $|\sum_k \sum_\omega \Phi^{\mathrm{int}}(x_k)_\omega - \Delta M|/|\Delta M|$ as a function of $n_\beta \in \{4, 8, 16, 32\}$.]**

C.4. ResNet-12 backbone

[Result: the metrics of Table 1 for the ResNet-12 backbone.]

C.5. Ablations

Table 3 removes one component of FAMA at a time.

Table 3. Ablations of FAMA on tieredImageNet→CUB (Conv4, 5-way 1-shot, 600 tasks). **[Placeholder values.]**

Variant	ABC$_\downarrow$	ABC$_\uparrow$	ρ_{reg}	Sign acc.
FAMA	[ ]	[ ]	[ ]	[ ]
without feature-reuse contrast (Φ of the full branch only)	[ ]	[ ]	[ ]	[ ]
with ReLU (unsigned)	[ ]	[ ]	[ ]	[ ]
last step only ($t = T$)	[ ]	[ ]	[ ]	[ ]
detached A_k (backbone block of λ_t dropped)	[ ]	[ ]	[ ]	[ ]
Grad-CAM channel pooling instead of the mask derivative	[ ]	[ ]	[ ]	[ ]

C.6. Sanity checks

[Result: Spearman correlation with the unperturbed map under cascading and independent layer randomization, under data randomization, and under the three support interventions of Section 4.4, each with the paired change of ΔM.]

D. Extended Related Work

This appendix complements Section 2 with work omitted there for space.

Few-shot learning. Metric-based methods classify queries by comparing embeddings with the support set (Vinyals et al., 2016; Snell et al., 2017). They adapt without gradient steps and therefore lie outside the scope of FAMA (Appendix B.3). Among optimization-based methods, Meta-SGD learns per-parameter rates (Li et al., 2017), MAML++ stabilizes training (Antoniou et al., 2019), and iMAML replaces the unrolled inner loop with implicit differentiation (Rajeswaran et al., 2019). Appendix B.2 covers the update rules of the first two. Cross-domain benchmarks expose the limits of all these methods (Chen et al., 2019; Guo et al., 2020), and our cross-domain tasks build on them.

Saliency methods. Beyond the methods of Section 2.3, guided backpropagation (Springenberg et al., 2015), CAM (Zhou et al., 2016), SmoothGrad (Smilkov et al., 2017), LIME (Ribeiro et al., 2016), and RISE (Petsiuk et al., 2018) also explain one prediction of one fixed model. The deletion metrics, sanity checks, and planted signals that evaluate such explanations are adapted to adaptation trajectories in Section 4.

Training-data attribution. Representer points also attribute a prediction to training examples, through the last layer of a model trained to a stationary point (Yeh et al., 2018). Like influence functions, they assume a converged model, which fast adaptation does not provide.

Differentiating through optimization. Reverse-mode differentiation through training has been used to tune hyperparameters (Maclaurin et al., 2015), and implicit differentiation avoids unrolling when the inner problem converges (Rajeswaran et al., 2019). Both serve to train models, whereas FAMA explains a model that is already trained.

Test-time adaptation. Beyond test-time training, TENT adapts a model by minimizing the entropy of its predictions on a test batch (Wang et al., 2021). Its objective is a label-free proxy of the kind discussed in Section 6. Such methods share our question of whether an update helps, and the template of Section 3.4 applies whenever the update is differentiable (Appendix B.2).